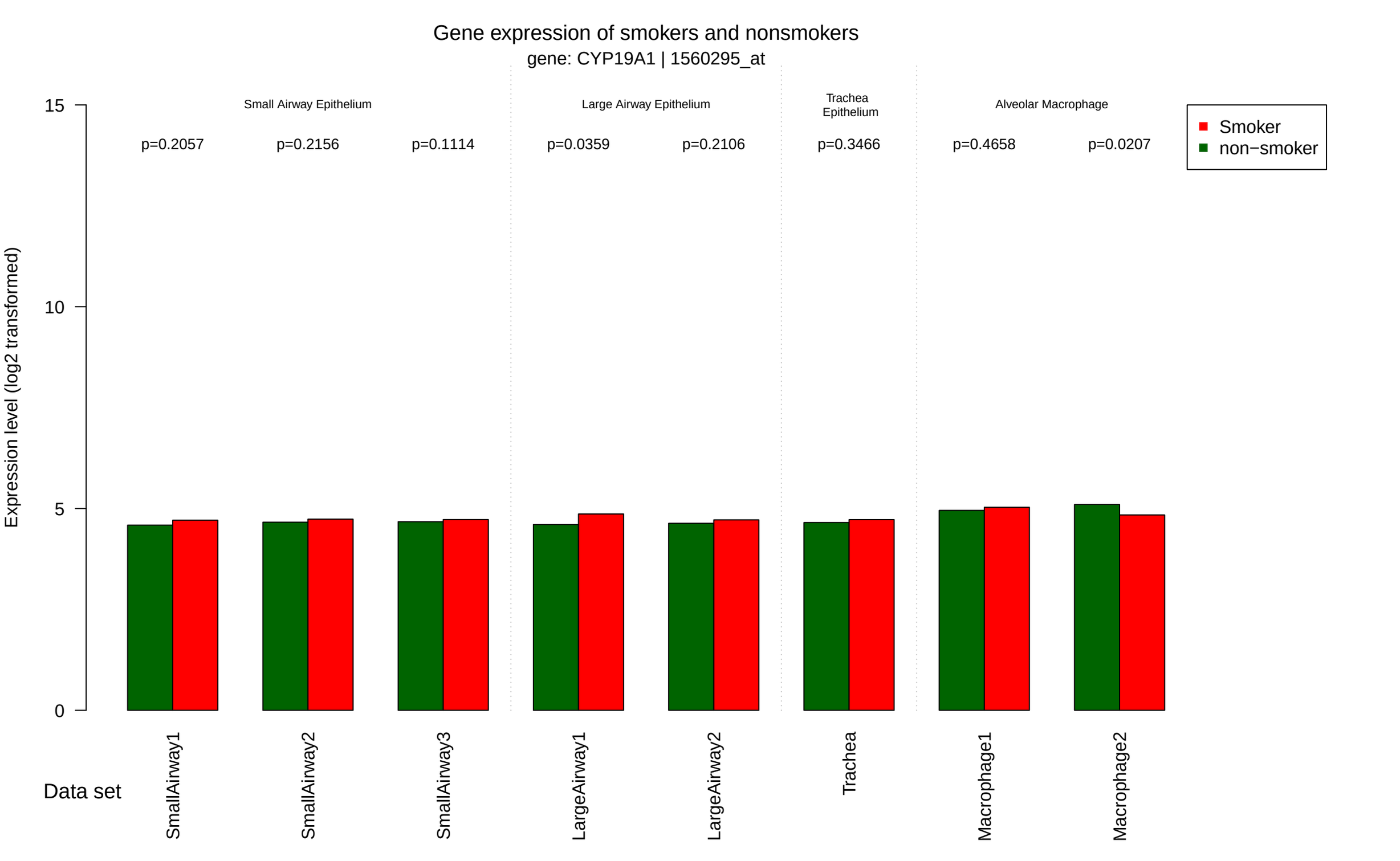
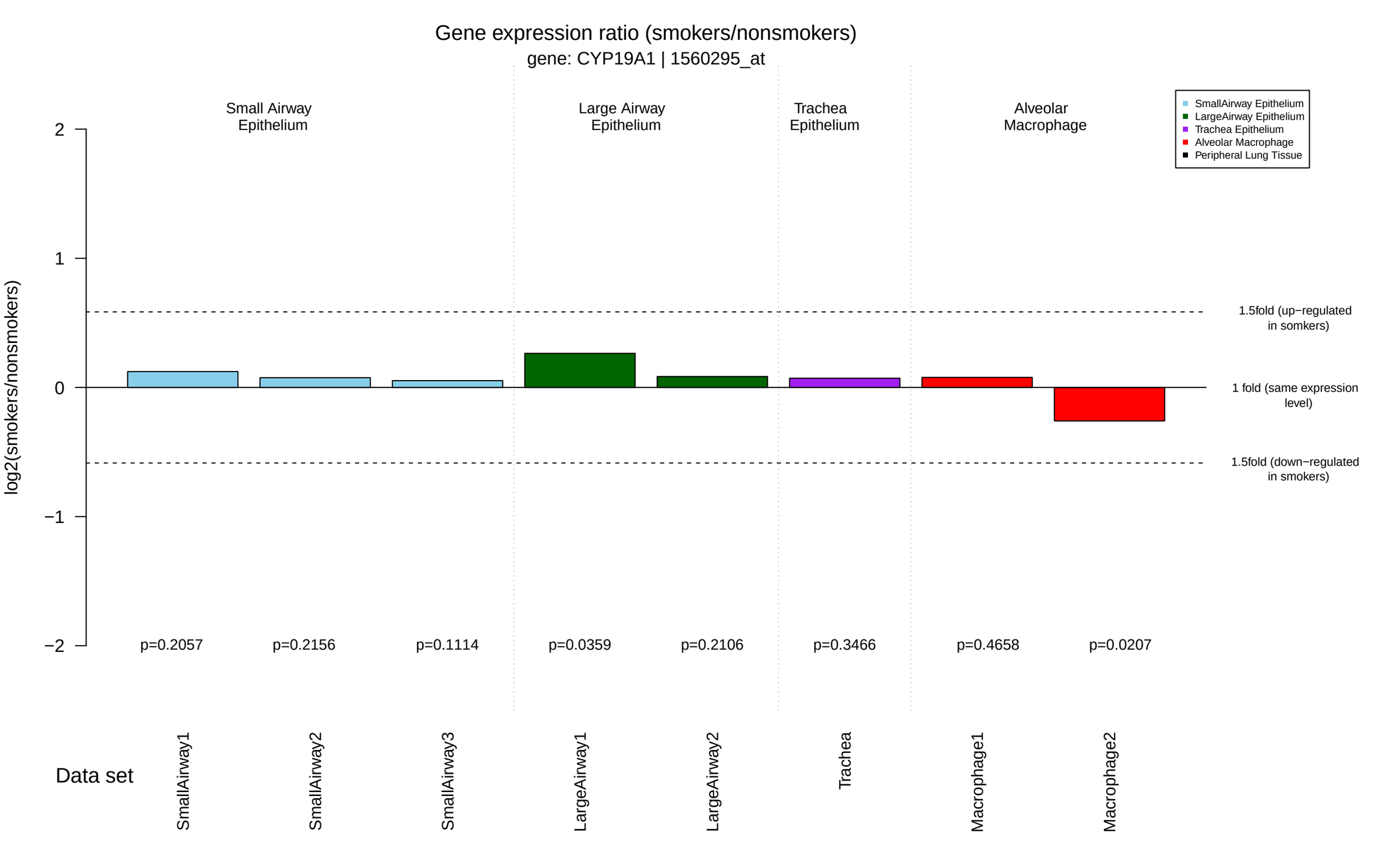
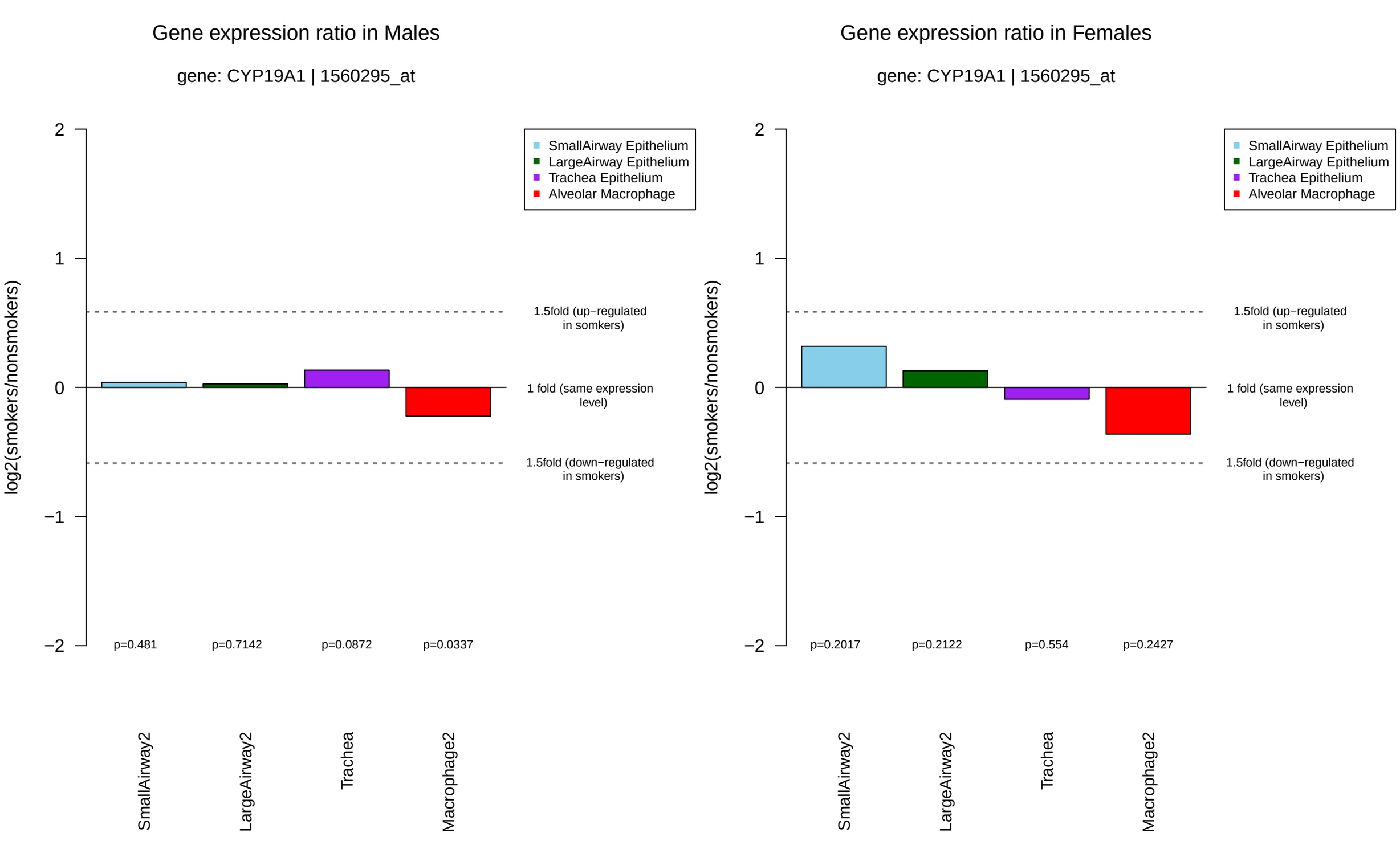
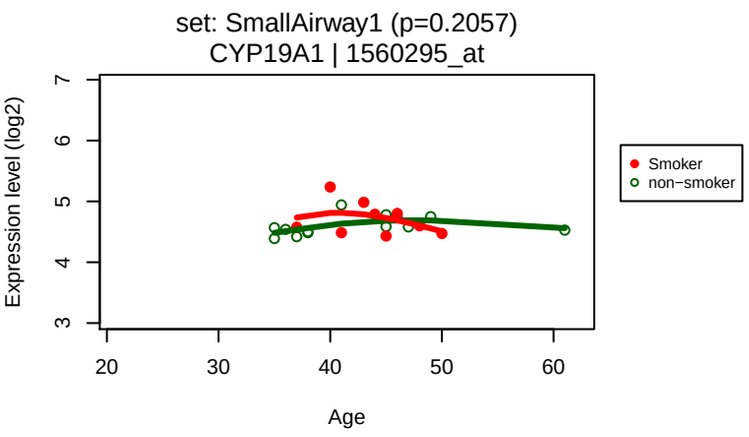
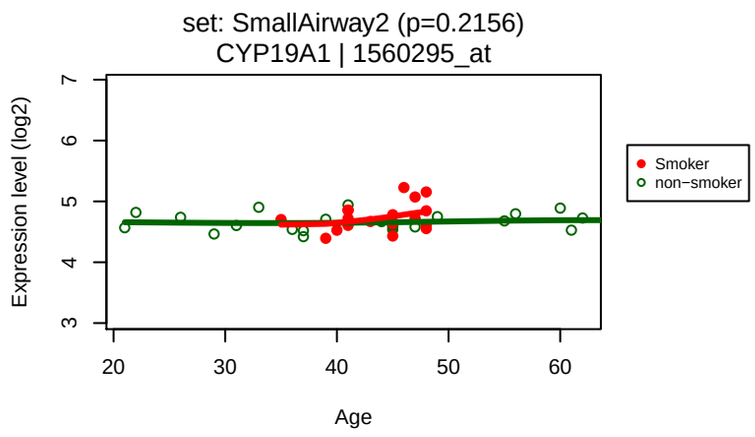
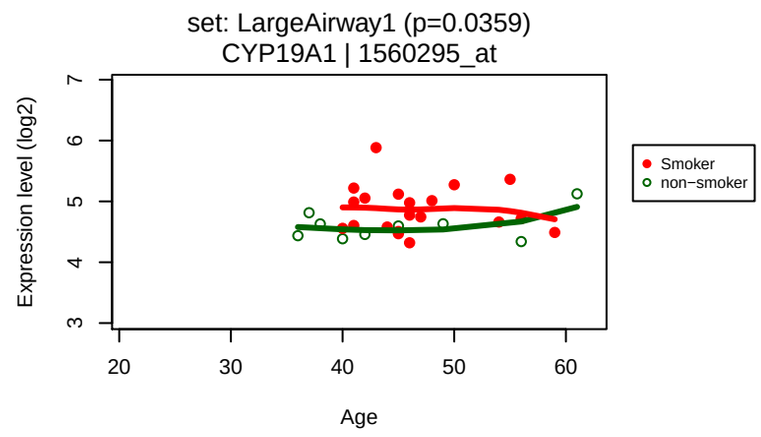

Gene expression of smokers and nonsmokers
gene: CYP19A1 | 1560295_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2057
p=0.2156
p=0.1114
p=0.0359
p=0.2106
p=0.3466
p=0.4658
p=0.0207
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

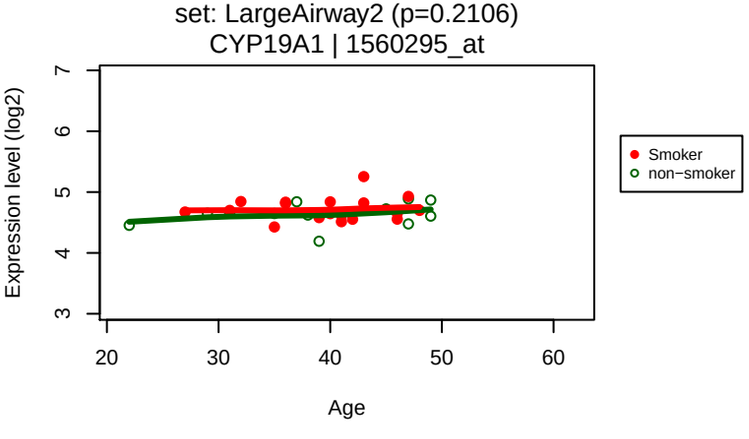

Gene expression ratio (smokers/nonsmokers)
gene: CYP19A1 | 1560295_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2057
p=0.2156
p=0.1114
p=0.0359
p=0.2106
p=0.3466
p=0.4658
p=0.0207
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

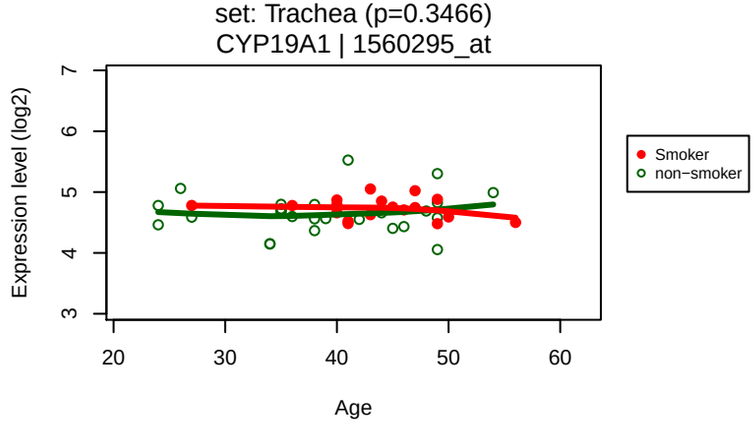

Gene expression ratio in Males
gene: CYP19A1 | 1560295_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.481
p=0.7142
p=0.0872
p=0.0337
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP19A1 | 1560295_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2017
p=0.2122
p=0.554
p=0.2427
SmallAirway2
LargeAirway2
Trachea
Macrophage2

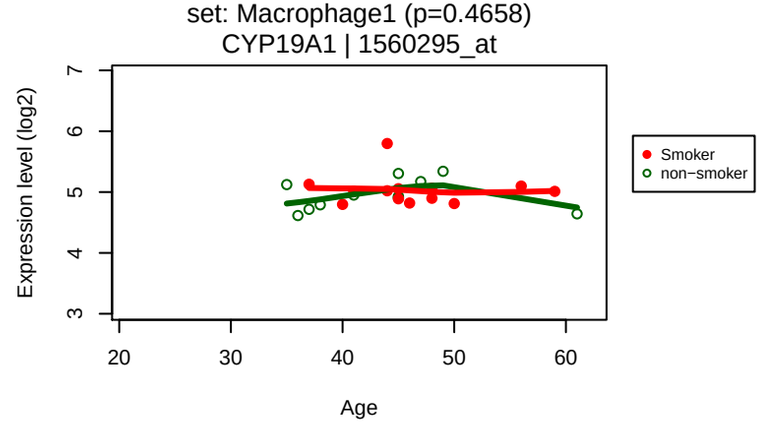

set: SmallAirway1 (p=0.2057)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker

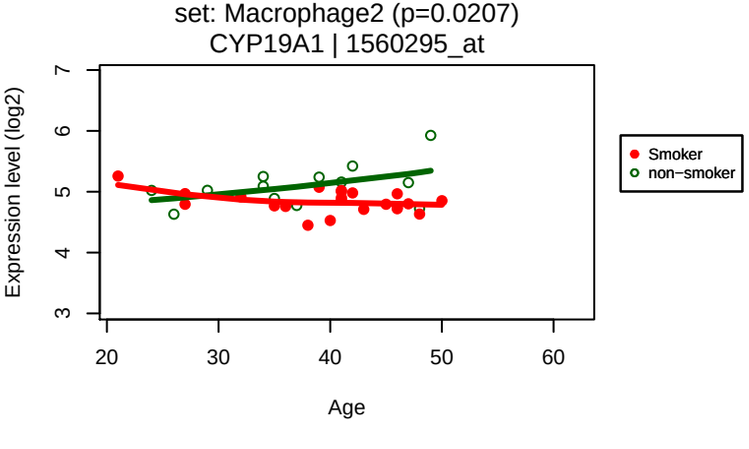

set: SmallAirway2 (p=0.2156)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker

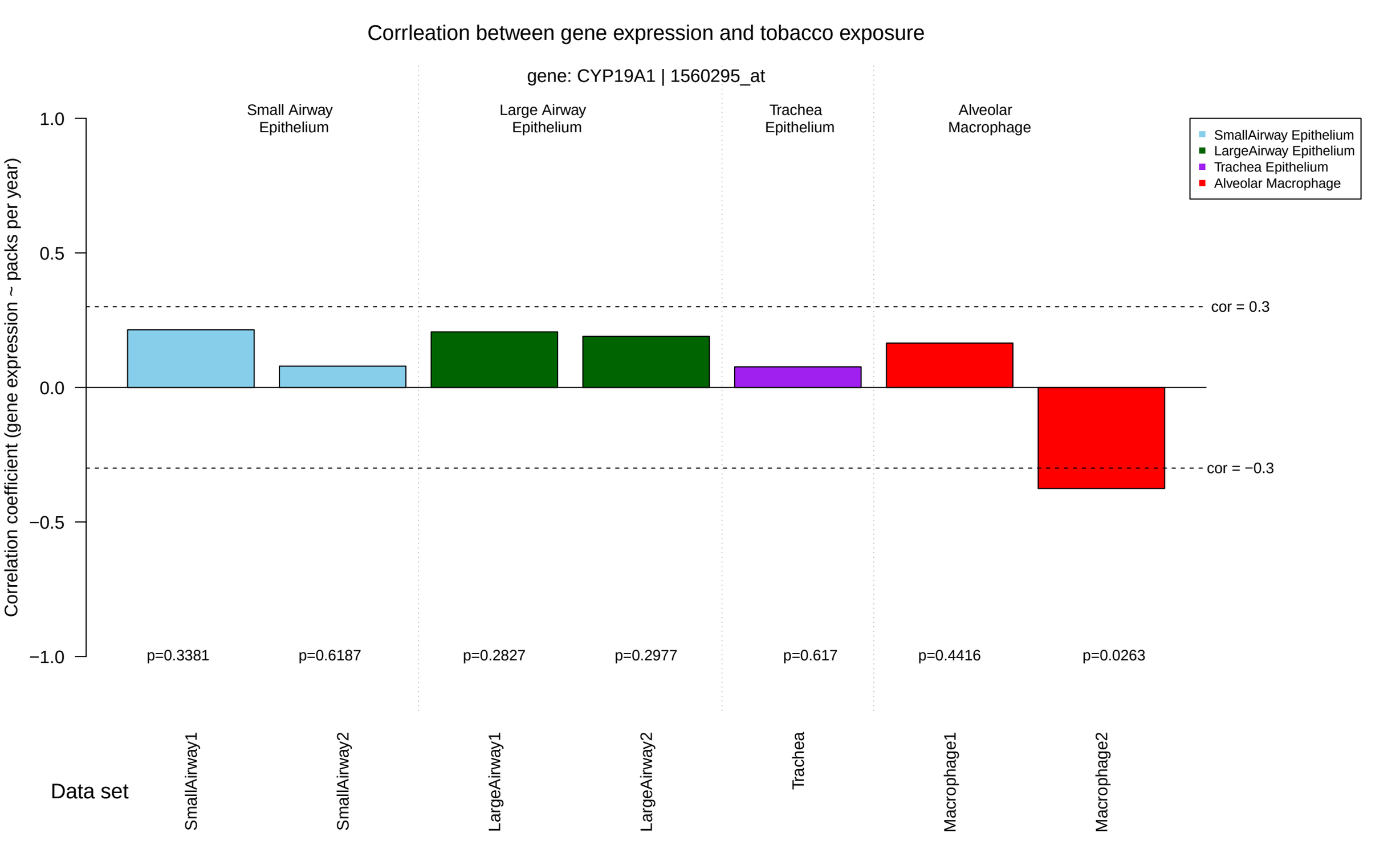

set: LargeAirway1 (p=0.0359)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2106)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3466)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4658)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0207)
CYP19A1 | 1560295_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP19A1 | 1560295_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3381
p=0.6187
p=0.2827
p=0.2977
p=0.617
p=0.4416
p=0.0263
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2